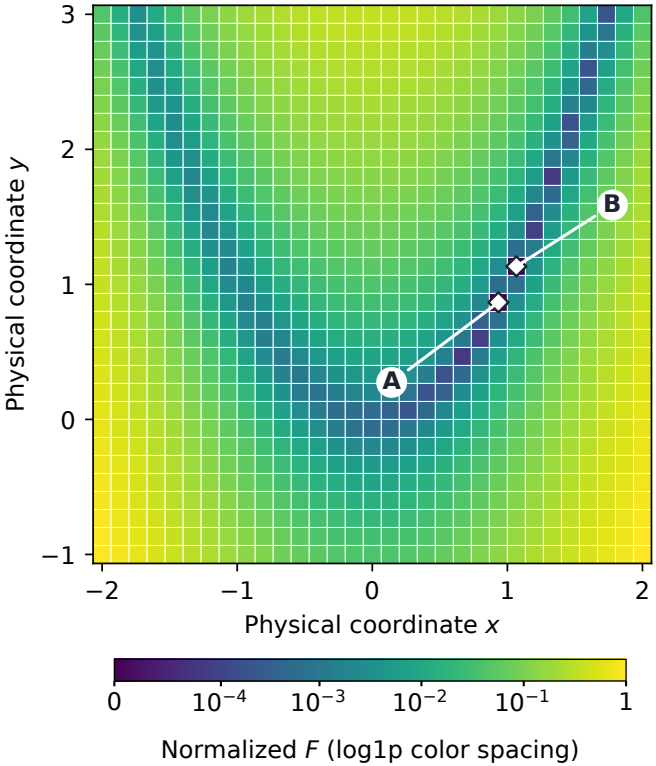
(a) Discrete objective landscape



(b) Coordinate-selection GDP

x-coordinate: select exactly one of 31

$$\bigvee_{i=0}^{30} \left[\; Y_i^x \;\; x = a_i = -2 + 4i/30 \; \right]$$

y-coordinate: select exactly one of 31

$$\bigvee_{j=0}^{30} \left[\; Y_j^y \;\; y = b_j = -1 + 4j/30 \; \right]$$

Both choices fix one physical point
$31 \times 31 = 961$ configurations; 62 Booleans

$$f(x,y) = (1-x)^2 + 100(y-x^2)^2$$
$$\min F(x,y) = (f(x,y) - L)/(U-L)$$
$$L = 13/2025, \quad U = 2509$$

Two exact lattice minima

A: $(i,j) = (22,14)$, $(x,y) = (14/15, 13/15)$

B: $(i,j) = (23,16)$, $(x,y) = (16/15, 17/15)$

Both: $f = 13/2025$ and $F = 0$ (exact).

Analytical objective at a fixed point.

Grid values shown without interpolation.